

Annex E – Data reported on MRSA from food-producing animals and derived meat

Annex to:

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E.1 Methicillin-resistant *Staphylococcus aureus* in food

Table 1: Methicillin-resistant *Staphylococcus aureus* in food, 2022.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Meat from bovine animals				
Netherlands	Fresh - processing plant monitoring	Batch	164	15 (9.1%)*
	Fresh - retail monitoring	Single	147	9 (6.1%)*
Meat from broilers				
Germany	Fresh - retail monitoring	Single	485	24 (4.9%)(a)
	Fresh - slaughterhouse monitoring	Batch	412	69 (16.7%)(b)
	Fresh - sampling at border control post (monitoring)	Single	49	0 (0%)
Netherlands	Fresh - retail monitoring	Single	186	14 (7.5%)*
Spain	Fresh - retail surveillance	Single	90	1 (1.1%)(c)
Meat from pig				
Netherlands	Fresh - retail monitoring	Single	180	13 (7.2%)*
Meat from turkey				
Germany	Fresh - retail monitoring	Single	460	158 (34.3%)(d)
	Fresh - slaughterhouse monitoring	Batch	417	224 (53.7%)(e)
Netherlands	Fresh - retail monitoring	Single	8	4 (50.0%)*
Meat from deer (venison)				
Netherlands	Fresh - sampling at border control post (monitoring)	Batch	15	0 (0%)
Meat from duck				
Germany	Fresh - retail monitoring	Single	372	7 (1.9%)*
	Fresh - slaughterhouse monitoring	Batch	350	3 (0.9%)*
Meat from farmed game - land mammals				
Netherlands	Fresh - sampling at border control post (monitoring)	Batch	2	1 (50.0%)*
Meat from wild game - birds				
Netherlands	Fresh - sampling at border control post (monitoring)	Batch	3	1 (33.3%)*
Meat from other animal species or not specified				

Netherlands	Fresh - sampling at border control post (monitoring)	Batch	1	1 (100%)*
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Fruits

Netherlands	Fresh - retail monitoring	Single	93	0 (0%)*
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^(a) *spa*-types: t011 (2), t034 (17), t571 (2), t2011 (1), t2330 (1), t10485 (1)

^(b) *spa*-types: t011 (32), t034 (22), t571 (1), t899 (1), t1430 (1), t1451 (1), t2011 (1), t6575 (1), *spa*-type not reported (9)

^(c) *spa*-types: t6228 (1)

^(d) *spa*-types: t008 (1), t011 (16), t034 (85), t127 (1), t899 (26), t1255 (1), t1422 (3), t1430 (3), t1580 (1), t2011 (2), t5452 (2), t10204 (2), *spa*-type not reported (15)

^(e) *spa*-types: t009 (1), t011 (23), t034 (142), t127 (4), t235 (1), t242 (2), t538 (1), t588 (2), t899 (20), t1255 (3), t1422 (2), t1430 (3), t1451 (1), t1793 (1), t2011 (5), t2346 (1), t2576 (1), t2922 (1), t10204 (1), t14089 (1), t21217 (1), CC398 *spa*-type not reported (2), *spa*-type not reported, (5)

**spa*-types not reported

Table 2: Methicillin-resistant *Staphylococcus aureus* in food, 2023.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Meat from bovine animals				
Austria	Fresh - retail monitoring	Single	310	6 (1.9%)(a)
Germany	Fresh - sampling at border control posts (monitoring)	Single	74	5 (6.8%)(b)
Netherlands	Fresh - processing plant monitoring	Batch	184	25 (13.6%)*
	Fresh - retail monitoring	Single	432	17 (3.9%)*
Meat from broilers				
Netherlands	Fresh - retail monitoring	Single	185	18 (9.7%)*
Meat from pigs				
Austria	Fresh - retail monitoring	Single	313	45 (14.4%)(c)
Germany	Fresh - retail monitoring	Single	380	39 (10.3%)(d)
Netherlands	Fresh - retail monitoring	Single	291	20 (6.9%)*
Slovakia	Fresh - retail monitoring	Single	130	20 (15.4%)*
Meat from turkey				
Netherlands	Fresh - retail monitoring	Single	7	2 (28.6%)*
Meat from deer (venison)				
Netherlands	Fresh - sampling at border control posts (monitoring)	Batch	11	0 (0%)
Meat from duck				
Netherlands	Fresh - sampling at border control posts (monitoring)	Batch	3	0 (0%)
Meat from farmed game - land mammals				
Netherlands	Fresh - sampling at border control posts (monitoring)	Batch	3	0 (0%)
Meat from sheep				
Netherlands	Fresh - retail monitoring	Single	151	7 (4.6%)*
Meat from wild game - birds				
Netherlands	Fresh - sampling at border control posts (monitoring)	Batch	5	1 (20.0%)*
Prawns				
Germany	Fresh - primary production monitoring	Single	10	0 (0%)

^(a) *spa*-types: t011 (3), t034 (3)

^(b) *spa*-types: t002 (4), t008 (1)

^(c) *spa*-types: t011 (20), t034 (12), t899 (1), t1255 (1), t1430 (3), t1451 (2), t2011 (4), t5524 (1), CC398, ST398 *spa*-type not reported (1)

^(d) *spa*-types: t008 (1), t011 (10), t034 (7), t127 (1), t693 (1), t899 (8), t1419 (1), t1430 (2), t1451 (1), t1580 (1), t2011 (2), t3512 (1), t20072 (1), *spa*-type not reported (2)

**spa*-types not reported

E.2 Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations excluded

Table 3: Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations excluded, 2022.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Cattle (bovine animals)				
Netherlands	Veal calves (under 1 year of age), farm survey	Holding	173	44 (25.4%)*
Pigs				
Belgium	Sows, farm monitoring	Herd	179	81 (45.3%)(a)
	Fattening pigs, farm monitoring	Herd	180	144 (80%)(b)
Norway	Farm control and eradication programme	Herd	591	0 (0%)
Slovakia	Fattening pigs, slaughterhouse monitoring	Animal	48	6 (12.5%)*

(a)*spa*-types: t011 (46), t034 (16), t588 (1), t1451 (1), t1457 (1), t2011 (3), t3423 (1), t5104 (1), t6575 (1), t15528 (1), *spa*-type not reported (9)

(b)*spa*-types: t011 (63), t034 (26), t779 (1), t1255 (1), t1451 (1), , t1580 (2), t2011 (2), t3423 (1), CC398 ST398 *spa*-type not reported (1), *spa*-type not reported (46)

**spa*-types not reported

Table 4: Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations excluded, 2023.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Cattle (bovine animals)				
Netherlands	Dairy cows, farm monitoring	Animal	316	2 (0.6%)*
Switzerland	Cattle under 1 year of age, slaughterhouse monitoring	Animal	307	11 (3.6%)*
Fowl (gallus gallus)				
Belgium	Broilers, farm monitoring	Herd	96	1 (1.0%)(a)
	Laying hens, farm monitoring	Herd	33	0 (0%)
Pigs				
Germany	Breeding pigs, farm monitoring	Herd	351	91 (51.9%)(b)
Norway	Pigs, farm control and eradication programme	Herd	541	0 (0%)
Slovakia	Pigs, slaughterhouse monitoring	Animal	130	27 (20.8%)*
Switzerland	Fattening pigs, slaughterhouse monitoring	Animal	310	166 (53.5%)*
Sheep				
Netherlands	Meat production sheep, farm survey	Herd/flock	156	8 (5.1%)*
Turkeys				
Belgium	Fattening turkeys, farm monitoring	Herd/flock	27	5 (18.5%)(c)

(a) *spa*-types: t011 (1)

(b) *spa*-types: t011 (19), t034 (38), t108 (1), t899 (6), t1451 (2), t1456 (1), t2011 (6), t2922 (2), t3075 (1), t4652 (1), t6575 (3), t6608 (1), t10204 (1), t16666 (1), t19248 (1), *spa*-type not reported (7)

(c) *spa*-types: t011 (5)

**spa*-types not reported

E.3 Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations

Table 5: Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations, 2022.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Cattle (bovine animals)				
Italy	Cattle, suspect sampling on farm	Animal	2	2 (100%)*

* *spa*-types not reported.

Note: The isolation method used for detection of MRSA is not considered in the analysis.

E.4 Methicillin-resistant *Staphylococcus aureus* in non-food-producing animals, clinical investigations

Table 6: Methicillin-resistant *Staphylococcus aureus* in non-food-producing animals, clinical investigations, 2022.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Cats				
Netherlands	Suspect sampling on veterinary clinics	Animal	200	4 (2%)*
Dogs				
Netherlands	Suspect sampling on veterinary clinics	Animal	3340	6 (0.2%)*
Solipeds, domestic				
Netherlands	Horse, suspect sampling on farm	Animal	327	15 (4.6%)*

**spa*-types not reported.

Note: The isolation method used for detection of MRSA is not considered in the analysis.

Table 7: Methicillin-resistant *Staphylococcus aureus* in non-food-producing animals, clinical investigations, 2023.

Country	Production type/monitoring description (where specified)	Sample unit	Number	
			Units tested	Positive for MRSA (%)
Cats				
Netherlands	Suspect sampling on veterinary clinics	Animal	2883	3 (0.1%)*
Dogs				
Netherlands	Suspect sampling on veterinary clinics	Animal	7969	8 (0.1%)*
Solipeds, domestic				
Netherlands	Suspect sampling on veterinary clinics	Animal	1028	13 (1.3%)*

**spa*-types not reported.

Note: The isolation method used for detection of MRSA is not considered in the analysis.

E.5 *spa*-type characterization in methicillin-resistant *Staphylococcus aureus*

Table 8: Methicillin-resistant *Staphylococcus aureus* *spa*-type characterization, 2022.

Category	Country	Animal/food type	Sample type/unit	No. of isolates	Where reported							
					<i>spa</i> -type	PVL status/IEC genes	ST	CC	<i>mec</i> -gene	Inferred CC	LA, CA or HA	Inferred ST/CC & type
Food-producing animals	Belgium	Pigs - breeding animals	Herd, OFM	43/179	11	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				14/179	34	<i>sak, scn⁺</i>	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	588	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	1451	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	1457	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				3/179	2011	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	3423	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	5104	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	6575	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	15528	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				2/179	11	-	3706	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	34	-	7645	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	11	-	8149	398	<i>mecA</i>	-	LA	CC398/LA
				1/179	34	-	8151	398	<i>mecA</i>	-	LA	CC398/LA
		Fattening pigs	Herd, OFM	60/180	11	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				24/180	34	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/180	779	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/180	1255	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/180	1451	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				2/180	1580	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				2/180	2011	-	398	398	<i>mecA</i>	-	LA	CC398/LA

Food	Germany	Meat from broiler	Batch, carcase, SM	1/180	3423	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/180		-	398	398	<i>mecA</i>	-	LA	CC398/LA
				2/180	11	-	3706	398	<i>mecA</i>	-	LA	CC398/LA
				1/180	34	-	7645	398	<i>mecA</i>	-	LA	CC398/LA
				1/180	34	-	8152	398	<i>mecA</i>	-	LA	CC398/LA
				1/180	11	-	8150	-	<i>mecA</i>	398	LA	CC398/LA
			Single, fresh, RM	32/412	11	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				22/412	34	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/412	571	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/412	899	-	-	-	<i>mecA</i>	1/398*	LA	CC1/CC398/LA
				1/412	1430	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/412	1451	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/412	2011	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/412	6575	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/485	11	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				17/485	34	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/485	571	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/485	2011	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/485	2330	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/485	10485	-	-	-	<i>mecA</i>	398	LA	CC398/LA
		Meat from turkey	Batch, carcase, SM	1/417	538	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/417	2922	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/417	14089	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/417	21217	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/417		-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/417	242	-	5	-	<i>mecA</i>	5	HA	CC5/HA
				1/417		-	398	-	<i>mecA</i>	398	LA	CC398/LA
				1/417	235	-	8325	-	<i>mecA</i>	1	LA	CC1/LA
				1/417	9	-	-	-	<i>mecA</i>	8	CA/HA	CC8/CA/HA

				23/417	11	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				142/417	34	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				4/417	127	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/417	242	-	-	-	<i>mecA</i>	5	HA	CC5/HA
				2/417	588	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				20/417	899	-	-	-	<i>mecA</i>	1/398*	LA	CC1/CC398/LA
				3/417	1255	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/417	1422	-	-	-	<i>mecA</i>	692	LA	CC692/LA
				3/417	1430	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/417	1451	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/417	1793	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				5/417	2011	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/417	2346	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/417	2576	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/417	10204	-	-	-	<i>mecA</i>	1	LA	CC1/LA
			Single, fresh, RM	1/460	8	-	-	-	<i>mecA</i>	8	CA/HA	CC8/CA
				16/460	11	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				85/460	34	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/460	127	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				26/460	899	-	-	-	<i>mecA</i>	1/398*	LA	CC1/CC398/LA
				1/460	1255	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				3/460	1422	-	-	-	<i>mecA</i>	692	LA	CC692/LA
				3/460	1430	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/460	1580	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/460	2011	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/460	5452	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/460	10204	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/90†	6228	-	-	-	<i>mecA</i>	398	LA	CC398/LA
Spain	Meat from broiler	Single, fresh, RM										

*Hybrid strain, t899 can be associated with CC1 and CC398. OFM, on-farm monitoring; RM, retail monitoring; SM, slaughterhouse monitoring; CC, clonal complex; ST, sequence type; CA, community-associated; HA, hospital-associated; LA, livestock-associated; IEC, immune evasion cluster; PVL, Panton-Valentin lukocidin.

Note: Purple columns show CA/HA-MRSA, beige columns show HA-MRSA, dark turquoise columns show CC398 LA-MRSA, light green columns show non-CC398 LA-MRSA and orange columns show likely LA-MRSA.

For isolates where "inferred CC is stated, the CC is based on the *spa*-type or ST as reported in literature.

Details on the categorisation of *spa*-types can be found in the chapter on MRSA in the main part of the report (Section 6.3.3 Food and animals: result from molecular typing of isolates). A summary of the information presented in this table is also visualised in Figure 3 of the MRSA chapter in the main part of the report.

† One isolate from sows carried the *sak* and *scn* genes associated with IEC.

‡ Spain reported 45 samples from fresh meat from broiler with skin and 45 samples from fresh meat from broiler without skin. These are reported together in the table.

Table 9: Methicillin-resistant *Staphylococcus aureus* *spa*-type characterization, 2023.

Category	Country	Animal/food type	Sample type/unit	No. of isolates	Where reported							
					<i>spa</i> -type	PVL status/ IEC genes	ST	CC	<i>mecA</i> -gene	Inferred CC	LA, CA or HA	Inferred ST/CC & type
Food-producing animals	Belgium	Broilers	Flock, OFM	1/96	11	-	398	398	<i>mecA</i>	-	LA	CC398/LA
		Fattening turkeys	Flock, OFM	3/27	11	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				2/27	11	-	8731	398	<i>mecA</i>	-	LA	CC398/LA
	Germany	Breeding pigs		1/351	3075	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/351	6608	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				1/351	16666	-	398	398	<i>mecA</i>	-	LA	CC398/LA
				6/351	899	-	9		<i>mecA</i>	1	LA	CC1/LA
				19/351	11	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				38/351	34	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/351	108	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/351	1451	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/351	1456	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				6/351	2011	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				2/351	2922	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/351	4652	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				3/351	6575	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/351	10204	-	-	-	<i>mecA</i>	1	LA	CC1/LA
			Herd, OFM	1/351	19248	-	-	-	<i>mecA</i>	398	LA	CC398/LA
Food	Austria	Meat from cattle	Single, fresh, RM	2/310	11	-	398	398	<i>mecA</i>		LA	CC398/LA
				3/310	34	-	398	398	<i>mecA</i>		LA	CC398/LA
				1/310	11	-	-	398	<i>mecA</i>		LA	CC398/LA
		Meat from pigs	Single, fresh, RM	1/313	899	-	9	1	<i>mecA</i>		LA	CC1/LA
				3/313	1430	-	9	1	<i>mecA</i>		LA	CC1/LA
				19/313	11	-	398	398	<i>mecA</i>		LA	CC398/LA

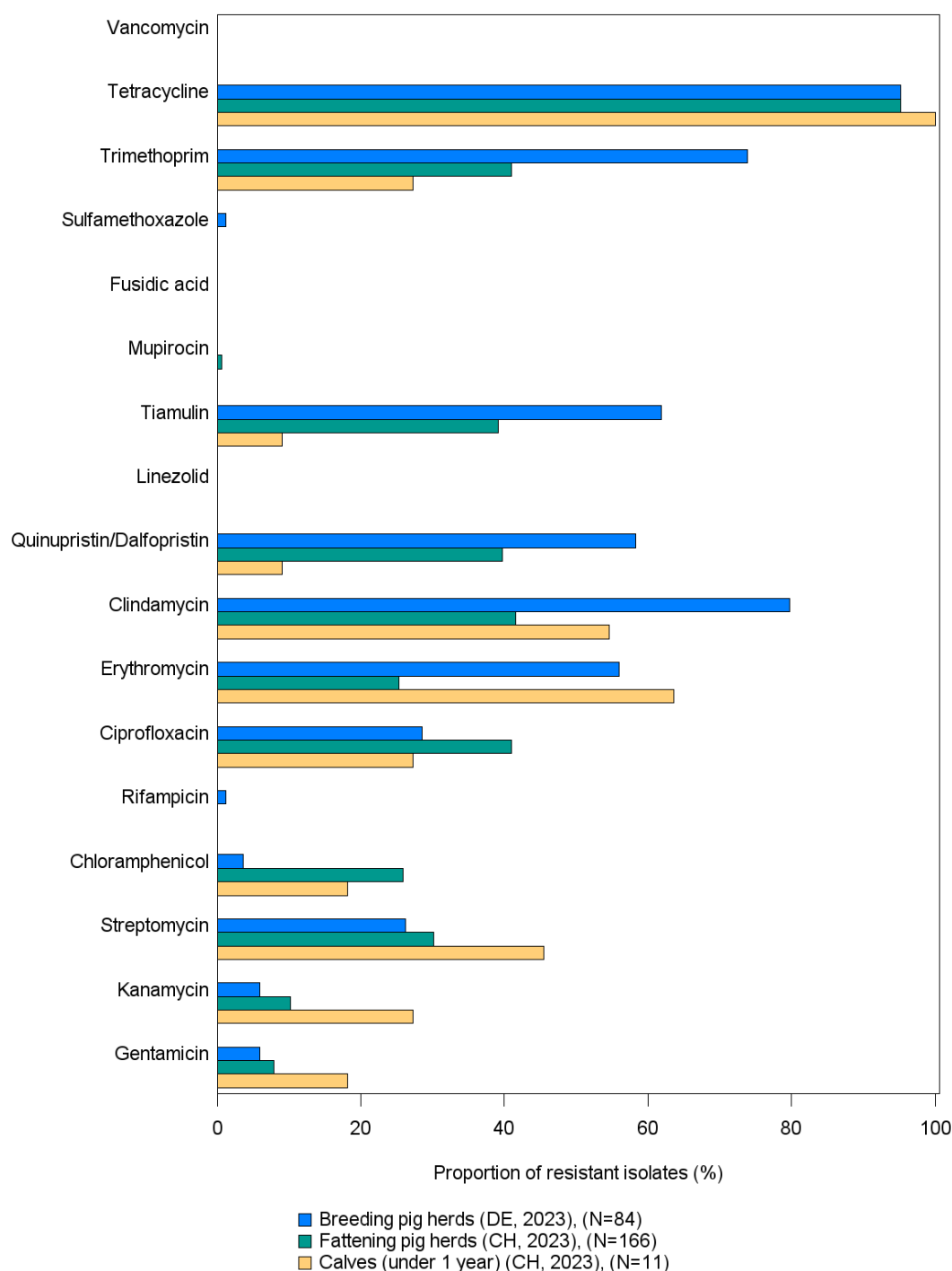
				12/313	34	-	398	398	<i>mecA</i>		LA	CC398/LA
				2/313	1451	-	398	398	<i>mecA</i>		LA	CC398/LA
				4/313	2011	-	398	398	<i>mecA</i>		LA	CC398/LA
				1/313	5524	-	398	398	<i>mecA</i>		LA	CC398/LA
				1/313	-	-	398	398	<i>mecA</i>		LA	CC398/LA
				1/313	11	-	-	398	<i>mecA</i>		LA	CC398/LA
				1/313	1255	-	-	398	<i>mecA</i>		LA	CC398/LA
	Germany	Meat from cattle	Single, fresh, BCP	4/74	2	-	-	-	<i>mecA</i>	5	HA	CC5/HA
				1/74	8	-	-	-	<i>mecA</i>	8	CA/HA	CC8/CA/HA
		Meat from pigs		1/380	693	-	9	-	<i>mecA</i>	1	LA	CC1/LA
				8/380	899	-	9	-	<i>mecA</i>	1	LA	CC1/LA
				1/380	20072	-	22	-	<i>mecA</i>	22	HA	CC22/HA
				1/380	8	-	-	-	<i>mecA</i>	8	CA/HA	CC8/CA/HA
				10/380	11	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				7/380	34	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/380	127	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/380	1419	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				2/380	1430	-	-	-	<i>mecA</i>	1	LA	CC1/LA
				1/380	1451	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/380	1580	-	-	-	<i>mecA</i>	398	LA	CC398/LA
			Single, fresh, RM	2/380	2011	-	-	-	<i>mecA</i>	398	LA	CC398/LA
				1/380	3512	-	-	-	<i>mecA</i>	1	LA	CC1/LA

*Hybrid strain, t899 can be associated with CC1 and CC398. OFM, on-farm monitoring; RM, retail monitoring; SM, slaughterhouse monitoring; BCP, border control post; CC, clonal complex; ST, sequence type; CA, community-associated; HA, hospital-associated; LA, livestock-associated; IEC, immune evasion cluster; PVL, Panton-Valentin lukocidin.

Note: Purple columns show CA/HA-MRSA, beige columns show HA-MRSA, dark turquoise columns show CC398 LA-MRSA and light green columns show non-CC398 LA-MRSA. For isolates where "inferred CC is stated, the CC is based on the *spa*-type or ST as reported in literature.

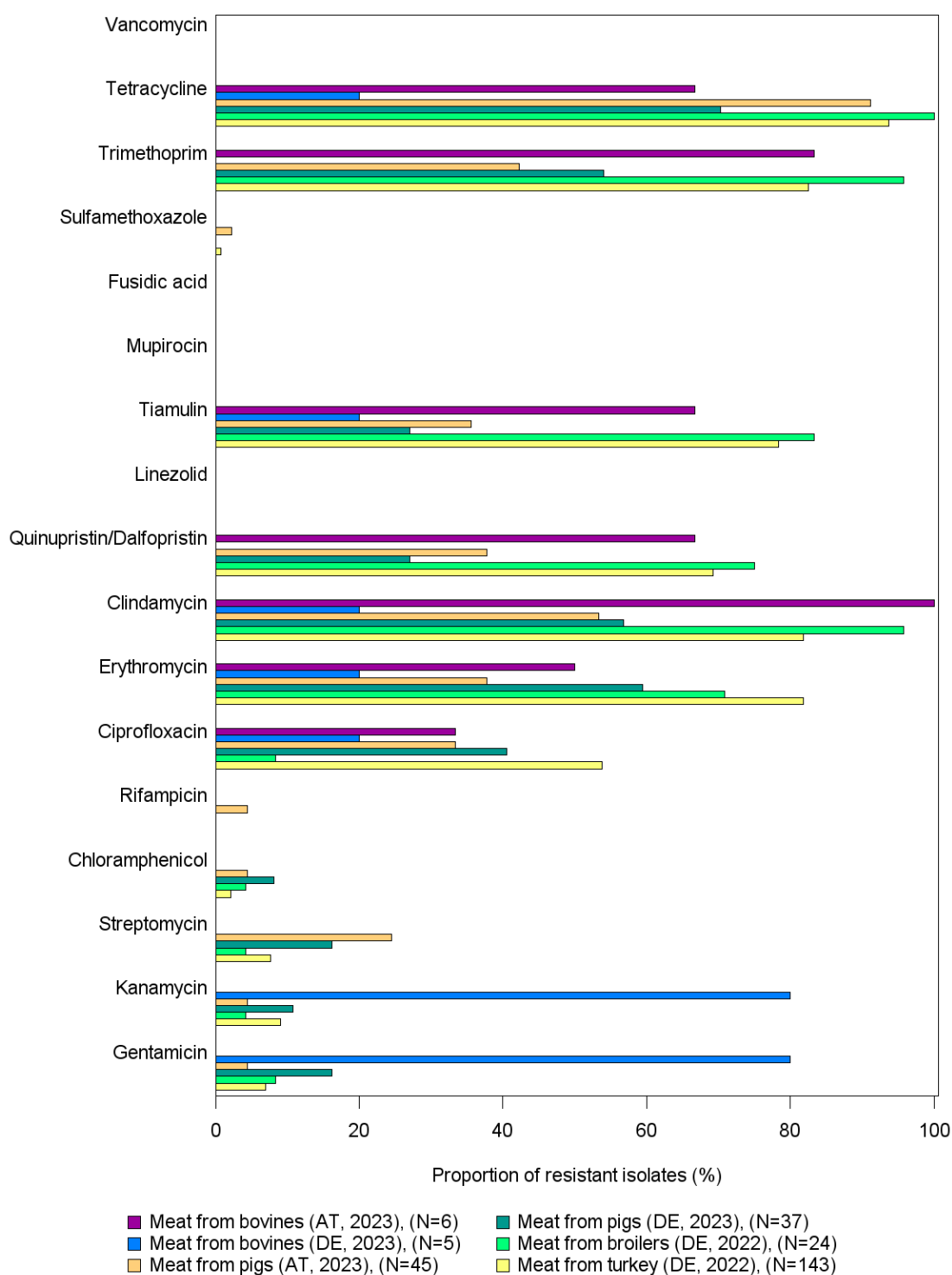
Details on the categorisation of *spa*-types can be found in the chapter on MRSA in the main part of the report (Section 6.3.3 Food and animals: Results of molecular typing of isolates). A summary of the information presented in this table is also visualised in Figure 60 of the MRSA chapter in the main part of the report.

E.6 Occurrence of resistance to selected antimicrobials in methicillin-resistant *Staphylococcus aureus*



CH : Switzerland, DE : Germany. All isolates were resistant to penicillin and ceftiofur.

Figure 1: Antimicrobial resistance in methicillin-resistant *Staphylococcus aureus* from animals, 2022 and 2023.



AT: Austria, DE: Germany. All isolates were resistant to penicillin and ceftiofur.

Figure 2: Antimicrobial resistance in methicillin-resistant *Staphylococcus aureus* from food, 2022 and 2023.

Table 10: Occurrence of resistance (%) to selected antimicrobials in methicillin-resistant *Staphylococcus aureus* from food and animals, 2022.

Country	N	GEN	KAN	STR	CHL	RIF	CIP	ERY	CLI	Q/D	LZD	TIA	MUP	FUS	SMX	TMP	TET	VAN
Meat from broilers - fresh																		
Spain	1 ^(a)	0	0	0	0	-	100	100	100	0	0	0	0	0	0	100	100	0
Germany	24 ^(b)	8.3	4.2	4.2	4.2	0	8.3	70.8	95.8	75	0	83.3	0	0	0	95.8	100	0
Meat from broilers - carcass																		
Germany	60 ^(c)	11.7	0	8.3	0	1.7	10	85	95	75	0	76.7	0	0	1.7	88.3	96.7	0
Meat from turkey - fresh																		
Germany	143 ^(d)	7.0	9.1	7.7	2.1	0	53.9	81.8	81.8	69.2	0	78.3	0	0	0.7	82.5	93.7	0
Meat from turkey - carcass																		
Germany	219 ^(e)	6.4	5.9	9.1	0	0.5	38.4	79.5	80.8	69.4	0	79.0	0	0	1.8	77.6	92.2	0

N, number of isolates tested; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; LZD, linezolid; TIA, tiamulin; MUP, mupirocin; FUS, fusidic acid; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline; VAN, vancomycin. All MRSA isolates were resistant to penicillin and ceftiofur.

^(a) *spa*-types: t6228 (1)

^(b) *spa*-types: t011 (2), t034 (17), t571 (2), t2011 (1), t2330 (1), t10485 (1)

^(c) *spa*-types: t011 (32), t034 (22), t571 (1), t899 (1), t1430 (1), t1451 (1), t2011 (1), t6575 (1)

^(d) *spa*-types: t008 (1), t011 (16), t034 (85), t127 (1), t899 (26), t1255 (1), t1422 (3), t1430 (3), t1580 (1), t2011 (2), t5452 (2), t10204 (2)

^(e) *spa*-types: t009 (1), t011 (23), t034 (142), t127 (4), t235 (1), t242 (2), t538 (1), t588 (2), t899 (20), t1255 (3), t1422 (2), t1430 (3), t1451 (1), t1793 (1), t2011 (5), t2346 (1), t2576 (1), t2922 (1), t10204 (1), t14089 (1), t21217 (1), CC398 *spa*-type not reported (2)

Table 11: Occurrence of resistance (%) to selected antimicrobials in methicillin-resistant *Staphylococcus aureus* from food and animals, 2023.

Country	N	GEN	KAN	STR	CHL	RIF	CIP	ERY	CLI	Q/D	LZD	TIA	MUP	FUS	SMX	TMP	TET	VAN
Cattle under 1 year																		
Switzerland	11*	18.2	27.3	45.5	18.2	0	27.3	63.6	54.5	9.1	0	9.1	0	0	0	27.3	100	0
Pigs																		
Switzerland	166*	7.8	10.4	30.1	25.9	0	41.0	25.3	41.6	39.8	0	39.2	0.6	0	0	41.0	95.2	0
Germany	84 ^(a)	6.0	6.0	26.2	3.6	1.2	28.6	56.0	79.8	58.3	0	61.9	0	0	1.2	73.8	95.2	0
Meat from pigs - fresh																		
Austria	45 ^(b)	4.4	4.4	24.4	4.4	4.4	33.3	37.8	53.3	37.8	0	35.6	0	0	2.2	42.2	91.1	0
Germany	37 ^(c)	16.2	10.8	16.2	8.1	0	40.5	59.5	56.8	27.0	0	27.0	0	0	0	54.1	70.3	0
Meat from cattle under 1 year - fresh																		
Austria	6 ^(d)	0	0	0	0	0	33.3	50.0	100	66.7	0	66.7	0	0	0	88.3	66.7	0
Germany	5 ^(e)	80.0	80.0	0	0	0	20.0	20.0	20.0	0	0	20.0	0	0	0	0	20.0	0

N, number of isolates tested; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; LZD, linezolid; TIA, tiamulin; MUP, mupirocin; FUS, fusidic acid; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline; VAN, vancomycin. All MRSA isolates were resistant to penicillin and ceftiofur.

^(a) *spa*-types: t011 (19), t034 (38), t108 (1), t899 (6), t1451 (2), t1456 (1), t2011 (6), t2922 (2), t3075 (1), t4652 (1), t6575 (3), t6608 (1), t10204 (1), t16666 (1), t19248 (1)

^(b) *spa*-types: t011 (20), t034 (12), t899 (1), t1430 (3), t1451 (2), t2011 (4), t5524 (1), t1255 (1), CC398 *spa*-type not reported (1)

^(c) *spa*-types: t008 (1), t011 (10), t034 (7), t127 (1), t693 (1), t899 (8), t1419 (1), t1430 (2), t1451 (1), t1580 (1), t2011 (2), t3512 (1), t20072 (1)

^(d) *spa*-types: t011 (3), t034 (3)

^(e) *spa*-types: t002 (4), t008 (1)

**spa*-types not reported.

E.7 MDR patterns in methicillin-resistant *Staphylococcus aureus*

Table 12: MDR patterns in methicillin-resistant *Staphylococcus aureus* isolates from animals and food, 2022.

MDR patterns in MRSA isolates	Germany				Spain	Total
	Meat from turkey, carcass (N=219)	Meat form turkey, fresh (N=143)	Meat from broilers, carcass (N=60)	Meat form broilers, fresh (N=24)	Meat from broiler, fresh (N=1)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
PNC-FOX-ERY-CLI-Q/D-TIA-TMP-TET	78 (35.6)	44 (30.8)	31 (52.5)	13 (54.2)		166 (37.2)
PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET	47 (21.5)	40 (28.0)	2 (3.4)			89 (20.0)
PNC-FOX-CIP	12 (5.5)	7 (4.9)				19 (4.3)
PNC-FOX-ERY-CLI-TIA-TMP-TET	11 (5.0)	4 (2.8)				15 (3.4)
STR-PNC-FOX-ERY-CLI-Q/D-TIA-TMP-TET	7 (3.2)	2 (1.4)	2 (3.4)			11 (2.5)
PNC-FOX-TET	6 (2.7)	4 (2.8)	1 (1.7)			11 (2.5)
GEN-KAN-PNC-FOX-TMP-TET	5 (2.3)	2 (1.4)		1 (4.2)		8 (1.8)
PNC-FOX-CIP-TIA-TMP-TET	3 (1.4)	5 (3.5)				8 (1.8)
PNC-FOX-CLI-Q/D-TIA-TET	7 (3.2)		1 (1.7)			8 (1.8)
PNC-FOX-CLI-Q/D-TIA-TMP-TET			3 (5.1)	4 (16.7)		7 (1.6)
GEN-KAN-PNC-FOX-CIP-ERY-CLI-TMP-TET	3 (1.4)	3 (2.1)				6 (1.3)
PNC-FOX-CIP-ERY-CLI-TIA-TMP-TET	3 (1.4)	3 (2.1)				6 (1.3)

MDR patterns in MRSA isolates	Germany				Spain	Total
	Meat from turkey, carcass (N=219)	Meat form turkey, fresh (N=143)	Meat from broilers, carcass (N=60)	Meat form broilers, fresh (N=24)	Meat from broiler, fresh (N=1)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
GEN-PNC-FOX-ERY-CLI-Q/D-TIA-TMP-TET	2 (0.9)		3 (5.1)			5 (1.1)
PNC-FOX-ERY-CLI-TET	1 (0.5)	2 (1.4)	2 (3.4)			5 (1.1)
PNC-FOX-TIA-TET	3 (1.4)	2 (1.4)				5 (1.1)
GEN-PNC-FOX-ERY-CLI-TMP-TET	1 (0.5)		2 (3.4)	1 (4.2)		4 (0.9)
KAN-STR-PNC-FOX-ERY-TET	3 (1.4)	1 (0.7)				4 (0.9)
PNC-FOX-CIP-ERY-CLI-TMP-TET	2 (0.9)		1 (1.7)		1 (100)	4 (0.9)
PNC-FOX-CIP-TET	2 (0.9)	2 (1.4)				4 (0.9)
PNC-FOX-ERY-CLI-TMP-TET		1 (0.7)	2 (3.4)	1 (4.2)		4 (0.9)
PNC-FOX-CIP-ERY-CLI	1 (0.5)	1 (0.7)	1 (1.7)			3 (0.7)
PNC-FOX-CIP-ERY-CLI-TET	2 (0.9)	1 (0.7)				3 (0.7)
PNC-FOX-CLI-TIA-TMP-TET			1 (1.7)	2 (8.3)		3 (0.7)
STR-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET	1 (0.5)	1 (0.7)		1 (4.2)		3 (0.7)
STR-PNC-FOX-ERY-CLI-Q/D-TIA-TET	2 (0.9)	1 (0.7)				3 (0.7)
GEN-STR-PNC-FOX-ERY-CLI-TMP-TET			2 (3.4)			2 (0.4)
GEN-STR-PNC-FOX-ERY-CLI-Q/D-TIA-TMP-TET	1 (0.5)	1 (0.7)				2 (0.4)

MDR patterns in MRSA isolates	Germany				Spain	Total
	Meat from turkey, carcass (N=219)	Meat from turkey, fresh (N=143)	Meat from broilers, carcass (N=60)	Meat from broilers, fresh (N=24)	Meat from broiler, fresh (N=1)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TET		2 (1.4)				2 (0.4)
PNC-FOX-CIP-ERY-CLI-Q/D-TIA-SMX-TMP-TET	1 (0.5)	1 (0.7)				2 (0.4)
PNC-FOX-ERY-CLI-Q/D-TIA-TET		1 (0.7)	1 (1.7)			2 (0.4)
PNC-FOX-RIF-ERY-CLI-Q/D-TIA-TMP-TET	1 (0.5)		1 (1.7)			2 (0.4)
CHL-PNC-FOX-CIP-ERY-CLI-TET				1 (4.2)		1 (0.2)
CHL-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET		1 (0.7)				1 (0.2)
FOX-CIP-ERY-CLI-Q/D-TIA-SMX-TMP-TET	1 (0.5)					1 (0.2)
GEN-KAN-PNC-FOX-CIP-CLI-TMP-TET		1 (0.7)				1 (0.2)
GEN-KAN-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET		1 (0.7)				1 (0.2)
GEN-KAN-PNC-FOX-CIP-TMP-TET		1 (0.7)				1 (0.2)
GEN-KAN-PNC-FOX-ERY-CLI-TMP-TET		1 (0.7)				1 (0.2)
GEN-KAN-STR-PNC-FOX-CIP-SMX-TMP-TET	1 (0.5)					1 (0.2)

MDR patterns in MRSA isolates	Germany				Spain	Total
	Meat from turkey, carcass (N=219)	Meat from turkey, fresh (N=143)	Meat from broilers, carcass (N=60)	Meat from broilers, fresh (N=24)	Meat from broiler, fresh (N=1)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
GEN-PNC-FOX-ERY-CLI-Q/D-TIA-TET	1 (0.5)					1 (0.2)
KAN-STR-CHL-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET		1 (0.7)				1 (0.2)
KAN-STR-PNC-FOX-CIP		1 (0.7)				1 (0.2)
KAN-STR-PNC-FOX-CIP-CLI-Q/D-TIA-TMP-TET		1 (0.7)				1 (0.2)
KAN-STR-PNC-FOX-ERY	1 (0.5)					1 (0.2)
PNC-FOX-CIP-CLI-Q/D-TIA-TMP-TET	1 (0.5)					1 (0.2)
PNC-FOX-CIP-CLI-TIA-TET	1 (0.5)					1 (0.2)
PNC-FOX-CIP-ERY-CLI-SMX-TMP-TET			1 (1.7)			1 (0.2)
PNC-FOX-CIP-ERY-CLI-Q/D-TMP-TET		1 (0.7)	1 (1.7)			1 (0.2)
PNC-FOX-CIP-ERY-TMP-TET		1 (0.7)				1 (0.2)
PNC-FOX-CIP-SMX-TET	1 (0.5)					1 (0.2)
PNC-FOX-TMP-TET			1 (1.7)			1 (0.2)
PNC-FOX-ERY-Q/D	1 (0.5)					1 (0.2)
PNC-FOX-Q/D-TIA-TET	1 (0.5)					1 (0.2)
STR-CHL-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET		1 (0.7)				1 (0.2)

MDR patterns in MRSA isolates	Germany				Spain	Total
	Meat from turkey, carcass (N=219)	Meat form turkey, fresh (N=143)	Meat from broilers, carcass (N=60)	Meat form broilers, fresh (N=24)	Meat from broiler, fresh (N=1)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
STR-PNC-FOX-CIP	1 (0.5)					1 (0.2)
STR-PNC-FOX-CIP-CLI-Q/D-TIA-TMP-TET			1 (1.7)			1 (0.2)
STR-PNC-FOX-CIP-ERY-CLI-TET	1 (0.5)					1 (0.2)
STR-PNC-FOX-ERY-CLI-TIA-TMP-TET	1 (0.5)					1 (0.2)
STR-PNC-FOX-CIP-ERY-CLI-TMP-TET		1 (0.7)				1 (0.2)
STR-PNC-FOX-TET	1 (0.5)					1 (0.2)
PNC-FOX-ERY	1 (0.5)					1 (0.2)
Total MDR	219 (100)	143 (100)	59 (98.3)	24 (100)	1 (100)	446 (99.8)

MDR, multidrug resistant; N, total number of isolates tested; n, number of isolates with MDR pattern; FOX, cefoxitin; PNC, penicillin; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; TIA, tiamulin; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline.

Table 13: MDR patterns in methicillin-resistant *Staphylococcus aureus* isolates from animals and food, 2023.

MDR patterns in MRSA isolates	Austria		Germany			Switzerland		Total
	Meat from pigs (n=45)	Meat from cattle (n=6)	Meat from pigs (n=37)	Meat from cattle (n=5)	Breeding pigs (n=84)	Cattle under 1 year (n=11)	Fattening pigs (n=166)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
PNC-FOX-ERY-CLI-Q/D-TIA-TMP-TET	2 (4.4)		5 (13.5)		16 (19.0)	1 (9.1)	19 (11.4)	43 (12.2)
PNC-FOX-TET	9 (20.0)		2 (5.4)		6 (7.1)	1 (9.1)	14 (8.4)	32 (9.1)
CHL-PNC-FOX-CIP-TET							31 (18.7)	31 (8.8)
PNC-FOX-CLI-Q/D-TIA-TMP-TET	2 (4.4)	1 (16.7)			10 (11.9)		10 (6.0)	23 (6.5)
STR-PNC-FOX-ERY-CLI-Q/D-TIA-TMP-TET			2 (5.4)		4 (4.8)		14 (8.4)	20 (5.7)
STR-PNC-FOX-TET	2 (4.4)				1 (1.2)	1 (9.1)	14 (8.4)	18 (5.1)
PNC-FOX-CIP	4 (8.9)		7 (18.9)					11 (3.1)
PNC-FOX-CIP-ERY-CLI-TMP-TET		1 (16.7)	4 (10.8)		5 (6.0)	1 (9.1)		11 (3.1)
PNC-FOX-CIP-TET	2 (4.4)		1 (2.7)		1 (1.2)		7 (4.2)	11 (3.1)
PNC-FOX-CLI-Q/D-TIA-TMP		2 (33.3)					8 (4.8)	10 (2.8)
STR-PNC-FOX-CLI-Q/D-TIA-TMP-TET	3 (6.7)				5 (6.0)		1 (0.6)	9 (2.5)
GEN-KAN-PNC-FOX-TET						1 (9.1)	6 (3.6)	7 (2.0)
STR-PNC-FOX-CIP-CLI-Q/D-TIA-TMP-TET	1 (2.2)				2 (2.4)		4 (2.4)	7 (2.0)

MDR patterns in MRSA isolates	Austria		Germany			Switzerland		Total
	Meat from pigs (n=45)	Meat from cattle (n=6)	Meat from pigs (n=37)	Meat from cattle (n=5)	Breeding pigs (n=84)	Cattle under 1 year (n=11)	Fattening pigs (n=166)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
STR-PNC-FOX-CIP-TET							7 (4.2)	7 (2.0)
PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET	2 (4.4)	1 (16.7)			2 (2.4)		1 (0.6)	6 (1.7)
PNC-FOX-TMP-TET			2 (5.4)				4 (2.4)	6 (1.7)
PNC-FOX-ERY-CLI-Q/D-TIA-TET	1 (2.2)		1 (2.7)		3 (3.6)			5 (1.4)
STR-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET	1 (2.2)		1 (2.7)		2 (2.4)		1 (0.6)	5 (1.4)
GEN-KAN-PNC-FOX-ERY-CLI-TET					1 (1.2)		3 (1.8)	4 (1.1)
PNC-FOX-ERY-CLI-TET	1 (2.2)	1 (16.7)			1 (1.2)	1 (9.1)		4 (1.1)
GEN-KAN-PNC-FOX				3 (75.0)				3 (0.8)
GEN-KAN-PNC-FOX-ERY-CLI-TMP-TET			2 (5.4)		1 (1.2)			3 (0.8)
GEN-KAN-PNC-FOX-TMP-TET					1 (1.2)	1 (9.1)	1 (0.6)	3 (0.8)
GEN-PNC-FOX-ERY-CLI-TMP-TET	1 (2.2)		1 (2.7)		1 (1.2)			3 (0.8)
KAN-STR-CHL-PNC-FOX-CIP-CLI-Q/D-TIA-TET							3 (1.8)	3 (0.8)
PNC-FOX-CIP-TMP-TET	1 (2.2)						2 (1.2)	3 (0.8)
STR-CHL-PNC-FOX-CIP-TET							3 (1.8)	3 (0.8)
CHL-PNC-FOX-CIP-ERY-CLI-TET							2 (1.2)	2 (0.6)

MDR patterns in MRSA isolates	Austria		Germany			Switzerland		Total
	Meat from pigs (n=45)	Meat from cattle (n=6)	Meat from pigs (n=37)	Meat from cattle (n=5)	Breeding pigs (n=84)	Cattle under 1 year (n=11)	Fattening pigs (n=166)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
CHL-PNC-FOX-ERY-CLI-TMP-TET	1 (2.2)				1 (1.2)			2 (0.6)
CHL-PNC-FOX-TET	1 (2.2)		1 (2.7)					2 (0.6)
GEN-KAN-STR-PNC-FOX-ERY-CLI-TMP-TET	1 (2.2)				1 (1.2)			2 (0.6)
PNC-FOX-CIP-CLI-Q/D-TIA-TMP-TET					2 (2.4)			2 (0.6)
PNC-FOX-CIP-ERY-CLI			1 (2.7)		1 (1.2)			2 (0.6)
PNC-FOX-ERY-CLI-Q/D-TMP-TET	1 (2.2)				1 (1.2)			2 (0.6)
PNC-FOX-ERY-CLI-TMP-TET	1 (2.2)				1 (1.2)			2 (0.6)
PNC-FOX-TIA-TET					2 (2.4)			2 (0.6)
STR-CHL-PNC-FOX-CIP-ERY-CLI-TET						2 (18.2)		2 (0.6)
STR-PNC-FOX-CIP-ERY-CLI-TET	2 (4.4)							2 (0.6)
STR-PNC-FOX-ERY-CLI-TET					1 (1.2)	1 (9.1)		2 (0.6)
CHL-PNC-FOX-CIP-TMP					1 (1.2)			1 (0.3)
CHL-PNC-FOX-CIP-MUP-TET							1 (0.6)	1 (0.3)
GEN-CHL-PNC-FOX-ERY-CLI-TMP-TET			1 (2.7)					1 (0.3)
GEN-KAN-PNC-FOX-CIP-ERY-CLI-TIA-TET				1 (25.0)				1 (0.3)
GEN-KAN-PNC-FOX-Q/D-TIA-TET							1 (0.6)	1 (0.3)

MDR patterns in MRSA isolates	Austria		Germany			Switzerland		Total
	Meat from pigs (n=45)	Meat from cattle (n=6)	Meat from pigs (n=37)	Meat from cattle (n=5)	Breeding pigs (n=84)	Cattle under 1 year (n=11)	Fattening pigs (n=166)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
GEN-KAN-STR-CHL-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TET							1 (0.6)	1 (0.3)
GEN-PNC-FOX-CIP-TET							1 (0.6)	1 (0.3)
GEN-PNC-FOX-ERY-CLI-TET			1 (2.7)					1 (0.3)
GEN-STR-PNC-FOX-ERY-CLI-TMP-TET			1 (2.7)					1 (0.3)
KAN-CHL-PNC-FOX-CIP-TET							1 (0.6)	1 (0.3)
KAN-PNC-FOX-ERY			1 (2.7)					1 (0.3)
KAN-STR-CHL-PNC-FOX-ERY-CLI-Q/D-TIA-TET			1 (2.7)					1 (0.3)
KAN-STR-PNC-FOX-CIP-CLI-Q/D-TIA-TMP-TET							1 (0.6)	1 (0.3)
KAN-STR-PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TMP-TET	1 (2.2)							1 (0.3)
KAN-STR-PNC-FOX-CIP-TMP-TET					1 (1.2)			1 (0.3)
KAN-STR-PNC-FOX-ERY-TET						1 (9.1)		1 (0.3)
PNC-FOX-CIP-CLI-TMP-TET							1 (0.6)	1 (0.3)
PNC-FOX-CIP-ERY-CLI-Q/D-TIA-TET	1 (2.2)							1 (0.3)
PNC-FOX-CIP-ERY-CLI-TET					1 (1.2)			1 (0.3)

MDR patterns in MRSA isolates	Austria		Germany			Switzerland		Total
	Meat from pigs (n=45)	Meat from cattle (n=6)	Meat from pigs (n=37)	Meat from cattle (n=5)	Breeding pigs (n=84)	Cattle under 1 year (n=11)	Fattening pigs (n=166)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
PNC-FOX-CIP-Q/D-TET							1 (0.6)	1 (0.3)
PNC-FOX-CIP-SMX					1 (1.2)			1 (0.3)
PNC-FOX-CLI-Q/D-TIA-TET	1 (2.2)							1 (0.3)
PNC-FOX-CLI-TIA-TMP-TET					1 (1.2)			1 (0.3)
PNC-FOX-ERY-TIA-TMP-TET							1 (0.6)	1 (0.3)
PNC-FOX-Q/D-TET							1 (0.6)	1 (0.3)
PNC-FOX-RIF-CIP-ERY-CLI-Q/D-TIA-TET					1 (1.2)			1 (0.3)
PNC-FOX-RIF-ERY-CLI-Q/D-TIA-TMP-TET	1 (2.2)							1 (0.3)
PNC-FOX-RIF-TET	1 (2.2)							1 (0.3)
PNC-FOX-SMX-TET	1 (2.2)							1 (0.3)
PNC-FOX-TIA-TMP-TET					1 (1.2)			1 (0.3)
PNC-FOX-TMP			1 (2.7)					1 (0.3)
STR-CHL-PNC-FOX-CIP-TMP-TET					1 (1.2)			1 (0.3)
STR-CHL-PNC-FOX-TET							1 (0.6)	1 (0.3)
STR-PNC-FOX-CIP			1 (2.7)					1 (0.3)
STR-PNC-FOX-CIP-ERY-CLI					1 (1.2)			1 (0.3)

	Austria		Germany			Switzerland		Total
	Meat from pigs (n=45)	Meat from cattle (n=6)	Meat from pigs (n=37)	Meat from cattle (n=5)	Breeding pigs (n=84)	Cattle under 1 year (n=11)	Fattening pigs (n=166)	
	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	n (% of MDR isolates)	
MDR patterns in MRSA isolates								
STR-PNC-FOX-CIP-ERY-CLI-TMP-TET					1 (1.2)			1 (0.3)
STR-PNC-FOX-CIP-TMP-TET					1 (1.2)			1 (0.3)
STR-PNC-FOX-ERY-CLI-Q/D-TIA-TET					1 (1.2)			1 (0.3)
Total MDR	45 (100)	6 (100)	37 (100)	4 (80)	84 (100)	11 (100)	166 (100)	353 (99.7)

MDR, multidrug resistant; N, total number of isolates tested; n, number of isolates with MDR pattern; FOX, ceftiofur; PNC, penicillin; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; TIA, tiamulin; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline.